Sri Lanka Institute of Information Technology

Faculty of Computing, Department of Computer Systems Engineering

IE3122: Network Security

Week 9 Tutorial Worksheet 2: IPsec, SSL VPN, and Cisco Configuration

Field	Detail
Based on	Week 9 Deck B, IPsec and SSL VPN lecture (IPsec's modular framework, AH/ESP, IKE phases, transport/tunnel mode, SSL VPN's functional blocks and client modes, IPsec vs SSL comparison, Cisco VPN configuration)
Learning Outcomes	LO3 (analyse security requirements), LO4 (evaluate security technologies), LO5 (configure security technologies)
Format	14 structured questions, no MCQs. Answer in your own words.
How to use this	Answer every question in the space provided. Together, your answers form a set of study notes for this topic. Use them directly when revising for the mid-semester test and final exam.

These questions are drawn directly from this week's IPsec and SSL VPN lecture. The final section's questions ask you to explain real Cisco configuration steps, which is deliberate: LO5 is specifically about configuring a security technology to satisfy a requirement, not just describing what it can do.

IPsec

Q1. Explain why IPsec is described as a modular framework rather than a single fixed protocol. Use the IPsec Protocol row specifically (AH, ESP, ESP+AH) to illustrate your answer. *[LO3, LO4]*

Q2. Explain the difference between AH and ESP, and explain why ESP or ESP+AH is used far more often than AH alone in real deployments. *[LO4]*

Q3. Explain the two phases of IKE negotiation, and explain why splitting negotiation into two phases is more efficient than negotiating everything at once, every time. *[LO4]*

Q4. Explain the difference between IPsec transport mode and tunnel mode, and explain which mode a site-to-site VPN would use and why. *[LO4]*

Q5. A network administrator has configured an IPsec tunnel using AH only, believing this protects the confidentiality of the data being sent. Evaluate this configuration and explain the security gap in the administrator's reasoning. *[LO3, LO4]*

Q6. Using the IPsec framework's Diffie-Hellman row, explain why choosing DH14 over DH2 is not simply a matter of preference. What would you tell an administrator who wanted to use the highest available Diffie-Hellman group on a piece of equipment that only supports up to Group 5? *[LO3, LO4]*

SSL VPN

Q7. Explain what a cipher suite is, and explain why TLS 1.3's approach to negotiating cipher suites is considered more tightly separated than TLS 1.2's. *[LO4]*

Q8. For each of the five functional blocks in SSL/TLS (protocol version, key exchange, bulk encryption, integrity/MAC, and identity/PKI), name one deprecated option and one currently recommended option. *[LO4]*

Q9. Explain the difference between clientless (portal) mode and full tunnel client mode in SSL VPN, and recommend which mode is more appropriate for a contractor who only needs to access one internal web-based timesheet application, briefly, twice a month. *[LO3, LO4]*

IPsec vs SSL VPN

Q10. Compare IPsec and SSL VPN specifically on NAT traversal and access granularity. Why does SSL VPN tend to have an easier time crossing NAT and firewalls than IPsec? *[LO4]*

Q11. A company wants remote employees to have full network-layer access to internal resources, equivalent to being physically in the office, while a separate group of external contractors should only reach one specific internal web application. Recommend which technology, IPsec or SSL VPN, and which SSL VPN mode if applicable, best fits each group, and justify both recommendations. *[LO3, LO4]*

Cisco Configuration

Q12. List the four configuration steps required to build a site-to-site IPsec VPN on a Cisco device, in the correct order, and briefly state what each step actually configures. *[LO1, LO5]*

Q13. Explain what the interesting traffic ACL in an IPsec configuration actually controls, and what happens to legitimate traffic that does not match this ACL. *[LO5]*

Q14. Explain what GRE over IPsec actually combines, and why the interesting-traffic ACL for a GRE-over-IPsec configuration matches the GRE traffic itself rather than the traffic riding inside the GRE tunnel. *[LO4, LO5]*

Turning This Into Exam Revision

Once you've answered every question, you have the beginnings of a proper study-notes set for this topic. To make it exam-ready:

1. Reread each answer and check it would make sense to you in three months, without the lecture slides open.
2. The IPsec configuration questions in the final section are close in spirit to what you will actually type in a lab. If you can answer them without looking at the slides, the hands-on work will go much faster.
3. Keep this worksheet with your notes for the mid-semester test (MCQ, LO1 and LO2) and final exam (MCQ + structured essay, LO1 through LO5).